

Facilitating Asymmetric Interaction between VR Users and External Users via Wearable Gesture-Based Interface

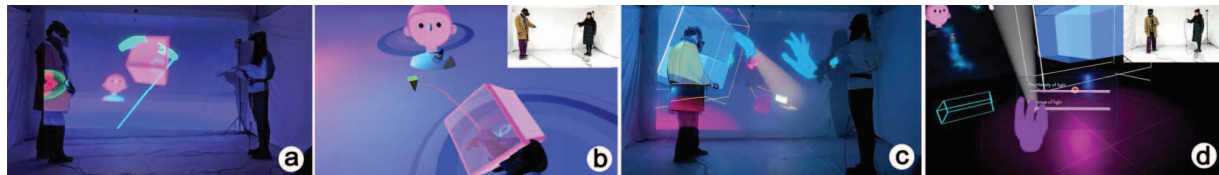
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Figure 1: The Gesture-based interaction (a)(c) and the controller-based interaction (b)(d) in scenario "Object Transition", "Collaborative Game" respectively.

ABSTRACT

Head-mounted displays (HMDs) create a highly immersive experience, while limit VR users' awareness of external users that co-located in the same environment. Previous studies generate various asymmetric communication that allowing external users to collaborate with VR users. In our work, we designed a wearable gesture-based interface for external users in two main asymmetric VR scenarios: (1) Object Transition and (2) Collaborative Game. We conducted a hybrid user study with twenty participants to compare the gesture-based interaction and controller-based interaction to investigate the effects on VR experience, social presence, and collaborative efficiency for external users. Our study revealed that the participants in gesture-based interaction received more positive feedbacks compared to those in controller-based interaction. Our findings suggested that the gesture-based interaction could be adapted for future asymmetric VR experience.

Keywords: Sharing VR experience, Asymmetric interaction, Gesture-based interaction.

Index Terms: Index Terms: Human-centered computing—Human computer interaction (HCI) — Interaction paradigms — Virtual reality

1 INTRODUCTION

Head-mounted displays provide the user with a highly immersive VR experience. However, the HMD separates VR users from the physical world, causing communication gaps with external users who co-locate at the same environment. Researchers present various interfaces that expand the virtual world into physical environment to generate asymmetric VR experience. Previous research on asymmetric VR experience mainly collaborates with VR users by using hand controllers and portable screen [1, 2]. However, these methods provide less immersion and limit the external users' interaction. In our study, we aimed to design a

novel interface that overcomes these shortages to enhance the external users' sense of immersion and their freedom of interaction. We propose a co-located VR experience in two main scenarios, (1) Object Transition and (2) Collaborative Game, in which we compared our gesture-based interaction and the traditional controller-based interaction for external users (See Figure 1). In a user study (N = 20), we compared the participants in both conditions. The gesture-based interaction showed significant advantages on VR experience, social presence, and collaborative efficiency.

2 SUPPORTING SYSTEM

The proposed system for gesture-based interaction is implemented in Unity 3D with off-the-shelf devices (HTC VIVE and Leap Motion). Figure 2 shows an overview of our system. The external user wears a VIVE tracker that attached to his or her head to locate the relative position in the virtual scenario together with the VR user. Meanwhile, an accompanying virtual camera projects the external user's perspective onto the physical environment. We use Leap Motion that tied around the external user's waist to enable hand gestures to interact with the VR user in real-time. The We encourage the external user to become a part of the shared VR experience and enable them to interact and explore the environment with the VR user in a collaborative way. For the traditional controller-based interaction, the external user uses a VIVE hand controller that attached a 7-inch removable screen as a virtual camera to display the scenario and perform the same tasks.

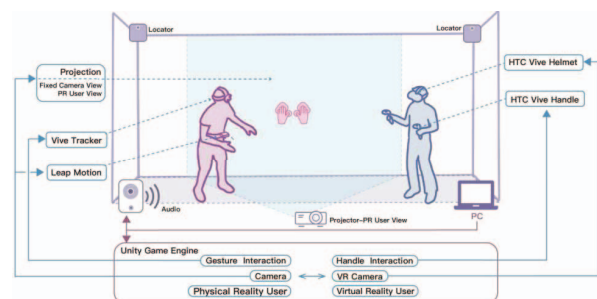


Figure 2: The proposed system for gesture-based interaction.

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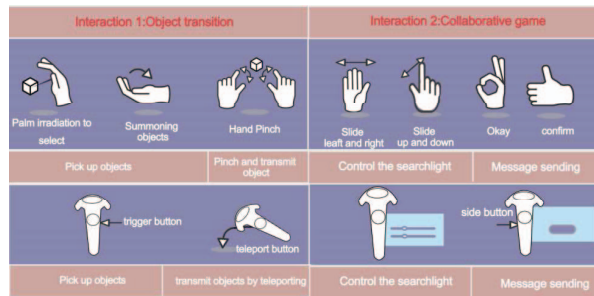


Figure 3: The instructions of gesture-based interaction and controller-based interaction in two scenarios.

3 OBJECT TRANSITION

The selection and transition of virtual objects are important interactive behaviours in VR collaboration [3, 4]. Therefore, we designed Object Transition. In this task, both VR and external users should pick up specific colour notes and throw them to each other. We provided object interaction by generating virtual rays in gesture-based interaction, which includes three hand gestures: *click*, *summon*, and *pinch* (see Figure 3). The external user can select the blue note through a generated blue ray by pinching with five fingers, then he or she can summon the note ball by flexing the four fingers and placed into the square to throw at the VR user by pinching hand (see Figure 1.a). The VR user can place the purple note into the square and pass it back to the external user as feedback. For the controller-based interaction, it also includes the interaction of picking up and transmitting the notes. The external user can press the handle button to pick the blue notes into the box and use the trigger button to teleport himself nearby the VR user (see Figure 1.b).

4 THE COLLABORATIVE GAME

The collaborative game is a typical interactive form in asymmetric VR [5]. Collaboration can be facilitated by empowering external users and VR users with different capabilities. Therefore, we designed an asymmetric cooperative game. There is a difference between using gesture-based interaction and controller-based interaction to send messages and control a searchlight (see Figure 1.c, d). For the gesture-base interaction, the external user looks for the location of the puzzles and send gestures to inform the VR user where the puzzles are. The VR user can teleport and pick up the puzzles in the virtual scenario only when the external user lights up the puzzles. We designed gestures: *slide*, *okay* and *confirm* (see Figure 3). The external user holds a searchlight in his or her left hand. Then he or she can control the range and brightness of the searchlight by pinching and sliding the index finger and thumb from the right hand and send an Okay gesture to inform the VR user that the puzzle has been found. VR users can confirm the message through the controller and teleport to pick up the puzzle. In this way, the external user can put their thumbs up and put the puzzle in the specified position. In the controller-based interaction, the external user can slide the slider to control the searchlight and click the UI buttons to send messages by using the controller.

5 PRIMARY STUDY

We conducted a primary study with 20 participants to study the VR experience, social presence and collaborative efficiency of the gesture-based interaction and the controller-based interaction. The participants were divided into ten groups (in pairs) and were randomly assigned as a VR user or an external user. All groups

tested both gesture-based interaction and controller-based interaction in two virtual scenarios. The participants need to fill the questionnaires of VR experience, social presence (NM-Mosp) and collaborative efficiency. Last, they were invited in a semi-structured interview.

We evaluated the interactions by using the 7-point Likert scale questionnaire and conducted the analysis by the Paired Samples t-test. The results showed that the VR experience for the participants in the gesture-based interaction was significantly higher than those in the controller-based interaction ($t(20) = 3.913, p = 0.004$). The gesture-based interaction provided external user with higher presence, involvement, and enjoyment. For social presence, the gesture-based interaction also reported significantly higher scores in comparison with the controller-based interaction ($t(20) = 2.790, p = 0.021$). For collaborative efficiency, in “Collaborative Game”, the gesture-based interaction reported significantly higher scores in comparison with the controller-based interaction ($t(20) = 2.516, p = 0.033$). External users use natural hand gestures to transmit information and cooperate more efficiently than the controller-based interaction. Due to the learning cost and accuracy rate of gesture, there were no significant differences in “Object Transition”.

6 CONCLUSION

In this paper, we conducted a hybrid user study in two scenarios by comparing the gesture-based interaction and the controller-based interaction in asymmetric VR. A primary study showed that our wearable gesture-based interface received more positive feedbacks on the sense of VR experience, social presence, and collaborative efficiency. The external user had a stronger sense of immersion and social experience due to the freedom of hand gestures and body perception. It could help both users perceive each other's interactions more clearly and give better feedback for collaboration. Meanwhile, our system still needs to be further improved in terms of learning cost and gesture design accuracy. The participants also provided additional suggestions for future implementation: 1) comparing gestures-based interaction and controller-based interaction in different collaborative scenarios, 2) designing more natural and cognitive gesture-based interaction.

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